Smart Parking System

C++ LOGIC ENGINE

10-Slot Real-Time Automated Allocation & Billing

10:36:48 pm

+ Park Vehicle

TOTAL SLOTS	AVAILABLE
10	8

OCCUPIED	REVENUE
2	Rs. 0

Park a Vehicle

Allocates first free slot (Slot 1-10)

Vehicle License Number

E.G. KA-05-MB-1234

Vehicle Type

Bike (Rs. 20/h) | Car (Rs. 40/h)

Assign Slot & Park

Remove / Checkout

Search plate or click any occupied slot

Enter Vehicle Number

E.G. KA-05-MB-1234

Calculate Fee & Checkout

Rate Calculation Rule (from C++ Model)

- Rate: Bike = **Rs. 20/hr**, Car/Other = **Rs. 40/hr**.
- Rule: `hours = max(1.0, ceil(difftime / 3600))`

Every fraction of an hour is rounded up to the nearest full hour, minimum 1 hour charged.

Parking Bay Status

10 Slots Total

Click any occupied slot to initiate instant checkout

Available Occupied

SLOT 1

OCCUPIED

KA-05-MH-2024

Car • Rs.40/h

45m 40s

Checkout >

SLOT 2

OCCUPIED

MH-12-BQ-8819

Bike • Rs.20/h

10m 40s

Checkout >

SLOT 3

AVAILABLE

Ready to Park

Status: Free

Slot 3

SLOT 4

AVAILABLE

Ready to Park

Status: Free

Slot 4

SLOT 5

AVAILABLE

Ready to Park

Status: Free

Slot 5

SLOT 6

AVAILABLE

Ready to Park

Status: Free

Slot 6

SLOT 7

AVAILABLE

Ready to Park

Status: Free

Slot 7

SLOT 8

AVAILABLE

Ready to Park

Status: Free

Slot 8

SLOT 9

AVAILABLE

Ready to Park

Status: Free

Slot 9

SLOT 10

AVAILABLE

Ready to Park

Status: Free

Slot 10

smart_parking_terminal.exe Clear

```
========================================
SMART PARKING SLOT SYSTEM
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[10:36:08 pm] Ready. 10 Parking Slots Online.
[10:36:08 pm] Awaiting vehicle arrivals or terminal commands...
[10:36:08 pm] Initial Demo vehicles loaded into Slot 1 and Slot 2.
```